

BioKEA Capabilities

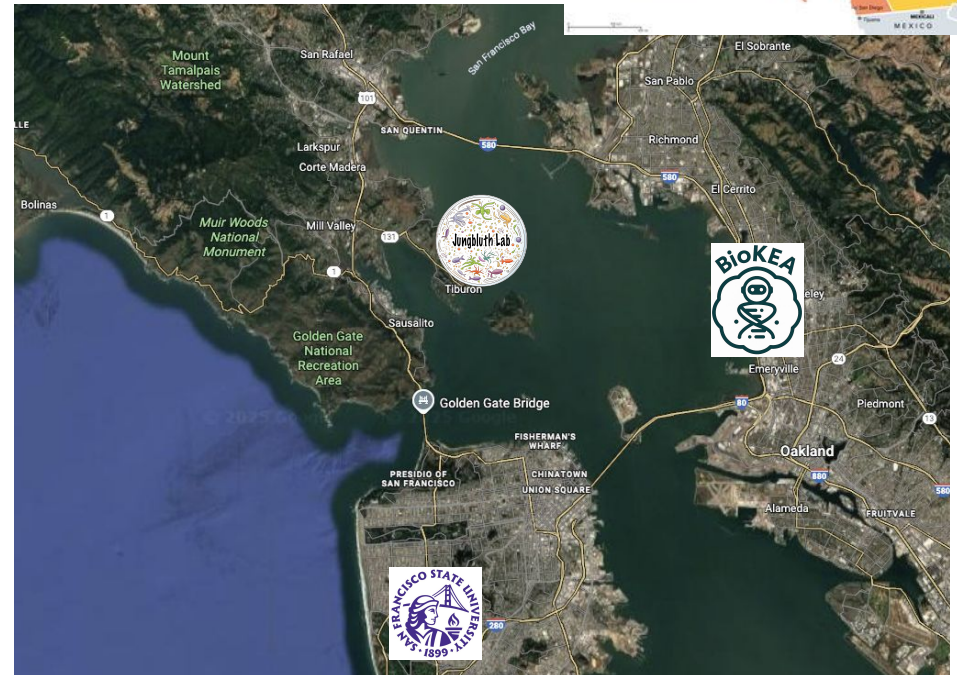
Jan 20, 2025



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BioKEA: Founded in March 2025

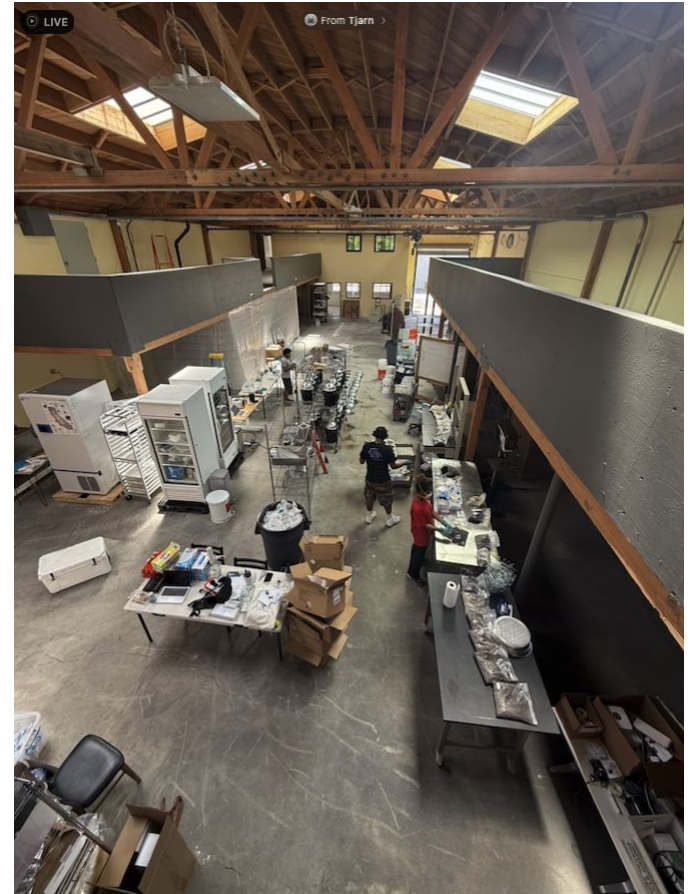
- **BioKEA: Biology Knowledge Exploration Assistant**
- **Spin-out:** Biodiversity Discovery + Environmental Omics Research in **San Francisco, USA**
- Leverages
 - **7+ years** at + DOE Berkeley Lab network
 - **7+ years** marine lab research
 - **5+ years** Bay Area environ. consulting
- Pursuit: **AI + genomics**
 - tools
 - datasets
 - publishing
- Built alongside a **shutdown SFSU** marine lab



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New Berkeley Labspace Capabilities

- BioKEA 7,000 square feet *free* warehouse space



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California Institute of Biodiversity Collaborative Expansion (Phase 1)

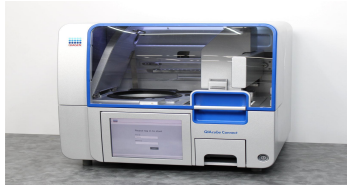
- **2x** KingFisher 96-well Extraction Robots
 - Auction, new in box; less than 1/10th retail cost
 - **Massive throughput** capabilities enable expansion

Extraction

Prep &
Amplification

Quantification

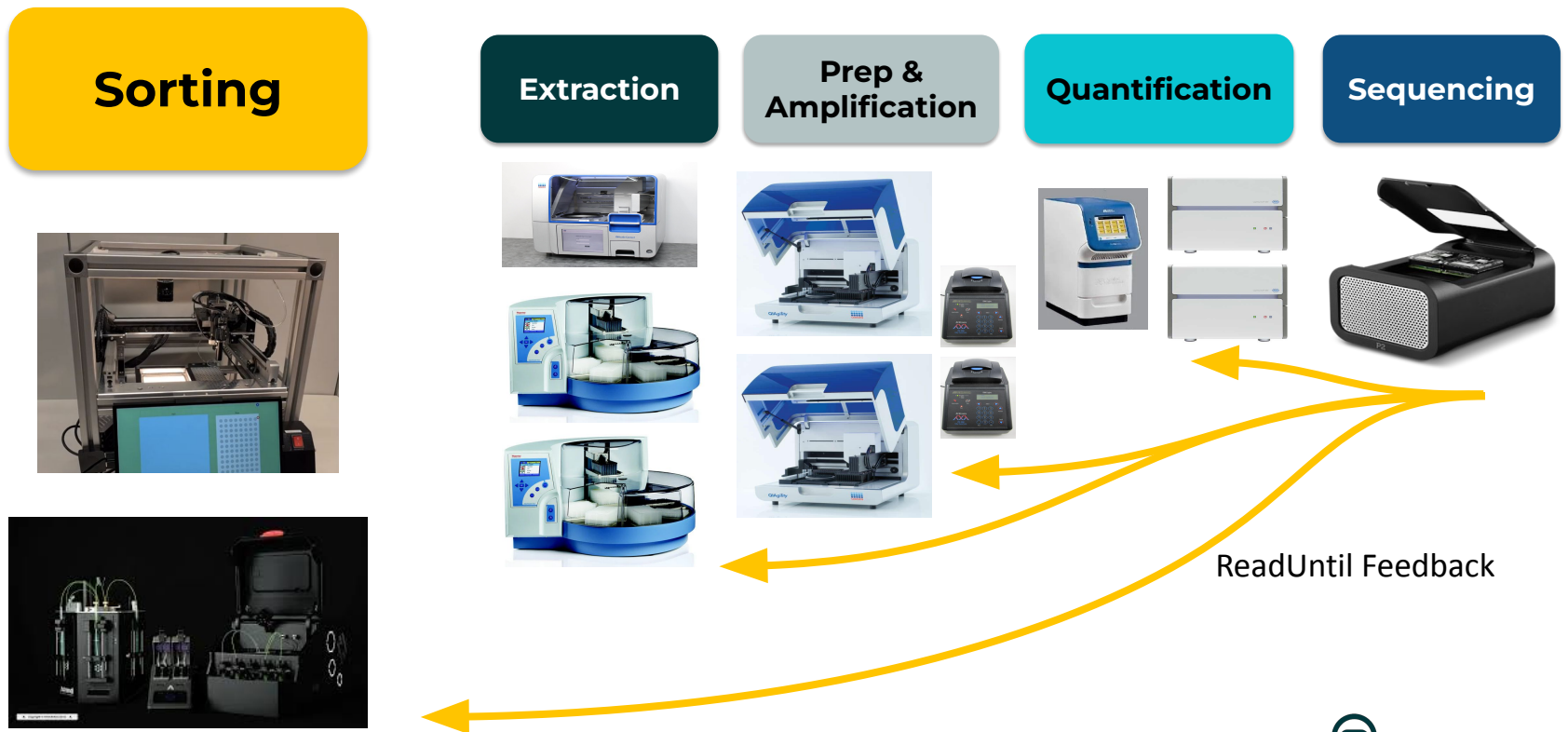
Sequencing



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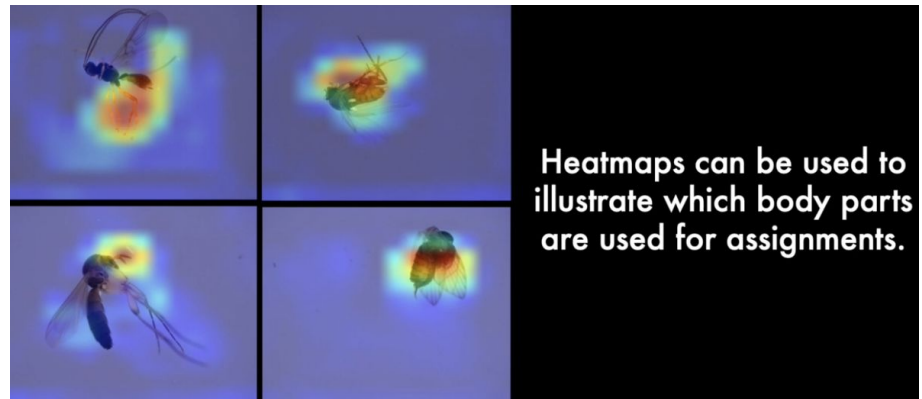
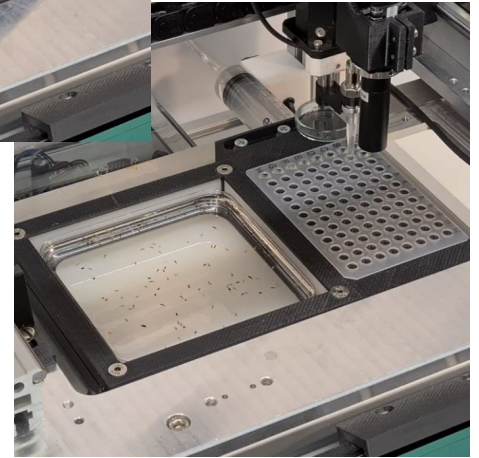
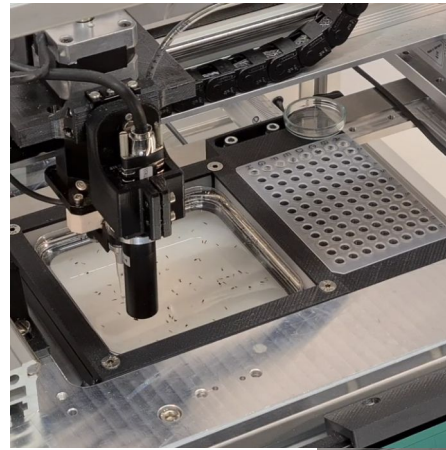
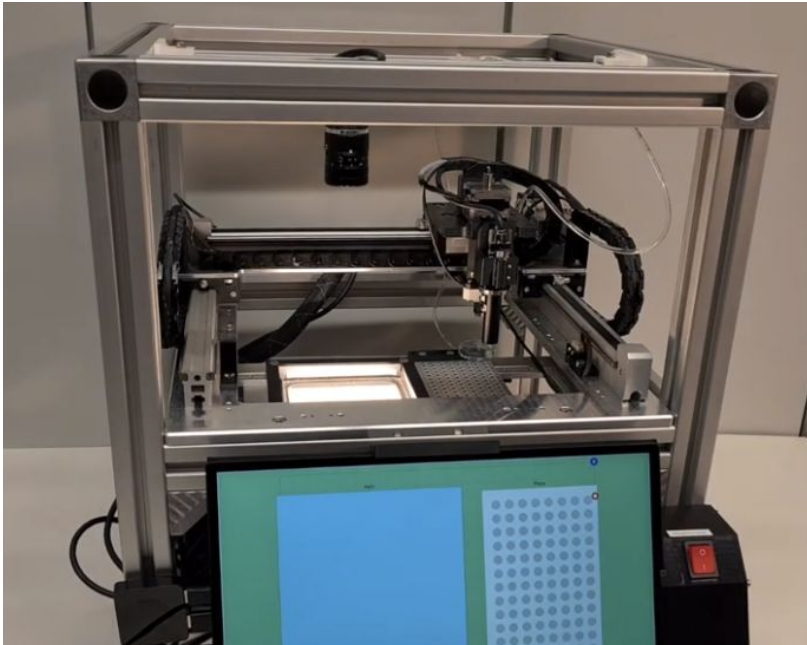
CIB Collaborative Expansion (Phase 2)

- Improved sample sorting capabilities
 - DNA extraction R&D
 - Sequencing feedback loop



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DiversityScanner: roboticized species discovery



Heatmaps can be used to
illustrate which body parts
are used for assignments.

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DiversityScanner: roboticized species discovery

Link: <https://www.youtube.com/watch?v=EIJ5VSHa4OI>

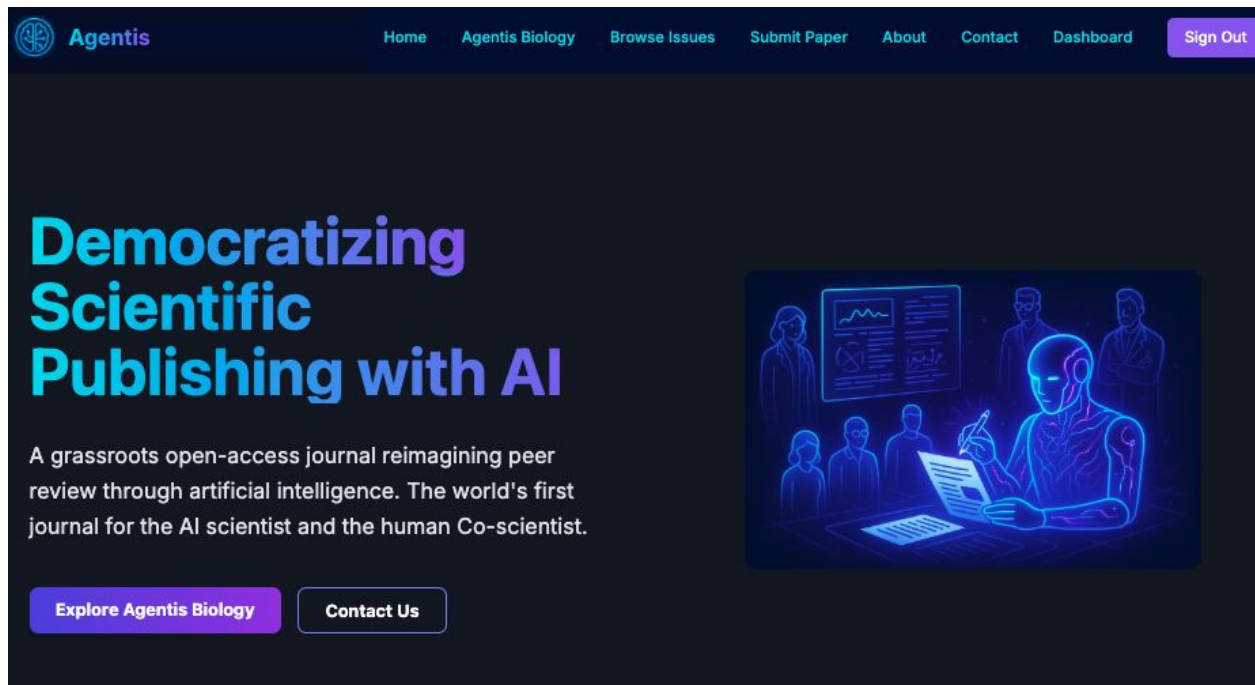
These are sample images of specimens.



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Science Storytelling & Changing Publication Standards

- Leverages unmet need for fast, reliable publishing of all results, positive and negative, in the AI era
- Accessible open science for everyone



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